

# MiniWorld 0

## Architecture schematics

### 1 The pair update

#### AlphaFold 3 · one operator, invoked twice [1, Alg. 12–13]

```

def TriangleMultiplication( $\{z_{ij}\}$ ,  $c = 128$ ,  $dir$ ):
    1:  $\hat{z}_{ij} = \text{LayerNorm}(z_{ij})$ 
    2:  $a_{ij}, b_{ij} = \sigma(\text{Linear}(\hat{z}_{ij})) \odot \text{Linear}(\hat{z}_{ij})$ 
    3:  $g_{ij} = \sigma(\text{Linear}(\hat{z}_{ij}))$ 
    4: if  $dir = \text{outgoing}$  then
    5:    $t_{ij} = \sum_k a_{ik} \odot b_{jk}$ 
    6: else
    7:    $t_{ij} = \sum_k a_{ki} \odot b_{kj}$ 
    8: end if
    9:  $\tilde{z}_{ij} = g_{ij} \odot \text{Linear}(\text{LayerNorm}(t_{ij}))$ 
    10: return  $\{\tilde{z}_{ij}\}$ 

```

$z_{ij} \in \mathbb{R}^{c_z}$   
 $a_{ij}, b_{ij} \in \mathbb{R}^c$   
 $g_{ij} \in \mathbb{R}^{c_z}$   
 Alg. 12  
 Alg. 13  
 $t_{ij} \in \mathbb{R}^c$

A pair block runs the whole body twice — once per direction, each call with its own parameters and its own residual:

$$z \mathrel{+}= \text{DropRow}_{0.25}(\text{TriMul}(z, \text{outgoing})), \quad z \mathrel{+}= \text{DropRow}_{0.25}(\text{TriMul}(z, \text{incoming}))$$

#### MiniWorld 0 · one bidirectional operator

```

def BidirectionalTriangleMultiplication( $\{z_{ij}\}$ ,  $h = c_z$ ):
    1:  $\hat{z}_{ij} = \text{LayerNorm}(z_{ij})$ 
    2:  $l_{ij} = \sigma(W_{lg} \hat{z}_{ij}) \odot W_l \hat{z}_{ij}$ 
    3:  $r_{ij} = \sigma(W_{rg} \hat{z}_{ij}) \odot W_r \hat{z}_{ij}$ 
    4:  $[l^{\rightarrow}, l^{\leftarrow}] = l_{ij}, [r^{\rightarrow}, r^{\leftarrow}] = r_{ij}$ 
    5:  $o_{ij}^{\rightarrow} = \sum_k l_{ik}^{\rightarrow} \odot r_{jk}^{\rightarrow}$ 
    6:  $o_{ij}^{\leftarrow} = \sum_k l_{ki}^{\leftarrow} \odot r_{kj}^{\leftarrow}$ 
    7:  $o_{ij} = \text{LayerNorm}([o_{ij}^{\rightarrow}, o_{ij}^{\leftarrow}])$ 
    8:  $g_{ij} = \sigma(W_g \hat{z}_{ij})$ 
    9: return  $z_{ij} + \text{DropRow}_p(g_{ij} \odot W_o o_{ij})$ 

```

$z_{ij} \in \mathbb{R}^{c_z}$   
 one shared input norm  
 $l_{ij} \in \mathbb{R}^{2h}$   
 $r_{ij} \in \mathbb{R}^{2h}$   
 each  $\in \mathbb{R}^h$   
 outgoing  
 incoming  
 one shared output norm  
 one shared output gate  
 residual fused in the epilogue

**Figure S1.** The triangular multiplicative update of AlphaFold 3, whose body is executed once per direction (*top*), and the single bidirectional operator of MiniWorld 0, which executes it once (*bottom*). Hidden width per direction and both contractions are unchanged; the input normalisation, output normalisation, output gate, dropout and residual are shared instead of duplicated. The residual on line 9 belongs to the operator by construction — it is folded into the kernel epilogue (§3) rather than added by the calling block.

## References

- [1] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, Sebastian W. Bodenstein, David A. Evans, Chia-Chun Hung, Michael O'Neill, David Reiman, Kathryn Tunyasuvunakool, Zachary Wu, Akvilė Žemgulytė, Eirini Arvaniti, Charles Beattie, Ottavia Bertolli, Alex Bridgland, Alexey Cherepanov, Miles Congreve, Alexander I. Cowen-Rivers, Andrew Cowie, Michael Figurnov, Fabian B. Fuchs, Hannah Gladman, Rishub Jain, Yousuf A. Khan, Caroline M. R. Low, Kuba Perlin, Anna Potapenko, Pascal Savy, Sukhdeep Singh, Adrian Stecula, Ashok Thillaisundaram, Catherine Tong, Sergei Yakneen, Ellen D. Zhong, Michal Zielinski, Augustin Židek, Victor Bapst, Pushmeet Kohli, Max Jaderberg, Demis Hassabis, and John M. Jumper. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, 630(8016):493–500, May 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-07487-w. URL <http://dx.doi.org/10.1038/s41586-024-07487-w>.